

YIMENG GU

guyimeng3@gmail.com ♦ <https://gymbeijing.github.io>

EDUCATION

Queen Mary University of London

Ph.D. in Computer Science

Advisor: Prof. Gareth Tyson

Research Interests: Multimodal Learning, MLLM/LLM Applications, MLLM/LLM Post-Training

London, UK

Sept 2021 - Nov 2025

Carnegie Mellon University

M.S. in Electrical and Computer Engineering

Pittsburgh, PA, USA

Aug 2020 - Aug 2021

The Hong Kong University of Science and Technology

MSc. in Big Data Technology

Hong Kong, China

Sept 2019 - June 2020

Beihang University

B.E. in Electronics Information Engineering

Shenyuan Honors College (top 50/3000+)

Beijing, China

Sept 2014 - June 2018

RESEARCH EXPERIENCE

Queen Mary University of London | PhD Student

London, UK

Multi-MLLM Knowledge Distillation for Out-of-Context News Detection

Aug 2024 - May 2025

- Designed a news evidence-based chain-of-thought (CoT) prompt to obtain predicted labels and rationales from multiple 70B-parameter teacher MLLMs as knowledge
- Distilled teachers' reasoning capability to a 7B student MLLM via two stages: LoRA fine-tuning for rapid absorption of aligned knowledge, then DPO to improve discernment on conflicting knowledge
- Improved detection performance by 13.5% with 10x fewer parameters and only 10% labeled data

Enhancing Fake News Detection with LLM-Augmented Reinforced Sampling

June 2024 - Feb 2025

- Designed a chain-of-thought (CoT) prompt to generate fake news in diverse styles from LLMs, enhancing the variety and quality of synthetic data
- Optimized the sampling ratio of fake to real news by modeling the task as a Markov Decision Process and leveraging Deep Q-learning for dynamic adjustment during training
- Improved the F1 score for fake news detection by 24.02% and 11.06% on benchmark datasets

Contrastive Domain Adaptation for Out-of-Context News Detection

Sept 2023 - Apr 2024

- Pioneered to explore domain adaptation in out-of-context news detection
- Developed a domain-invariant feature learning approach incorporating contrastive learning and maximum mean discrepancy (MMD), reducing discrepancies between target and source domains
- Integrated confidence-based test-time training (TTT) for further target domain adaptation
- Improved domain adaptation performance by 2.6% on benchmark datasets

WORK EXPERIENCE

Autodesk Research | Research Intern in AI Lab

London, UK

Generative AI for Identifying Mechanisms in Fusion360 Assemblies

Jun 2023 - Sept 2023

- Introduced a part-graph representation method by leveraging CLIP-based multi-view image encoding and TSP-guided ordering to transform part graphs into embedding sequences
- Proposed an AutoEncoder-latentGAN framework to first model the probabilistic distribution of a given part's neighboring parts in an assembly and then predict them
- Designed IoU-based and qualitative evaluation metrics; proposed method predicts neighbors of given parts effectively, improving IoU by 3% over baseline

- Annotated 360+ Chinese labor contracts to locate and extract key fields: *job position*, *job location*, *job description* and *part-time/full-time*
- Applied R-Net (reading comprehension) and BiLSTM-CRF (named entity recognition) to the task and achieved over 50% improvement in exact match (EM) and accuracy

ENGINEERING PROJECTS

The Design and Implementation of the Carcassonne Game

Mar 2021

- Designed system architecture and built a multiplayer Carcassonne game in Java (3.5k lines of code)
- Implemented interactive game interface with real-time tile placement and scoring using Java Swing
- Automated unit and integration testing with JUnit and Travis CI to validate core modules

SELECTED PUBLICATIONS

1. Coarse-to-Refined: Multi-MLLM Knowledge Distillation for Out-of-Context News Detection
Yimeng Gu, Zhao Tong, Ignacio Castro, Shu Wu, Gareth Tyson
In *IEEE ICME*, 2026
2. Generate First, Then Sample: Enhancing Fake News Detection with LLM-Augmented Reinforced Sampling
Zhao Tong*, **Yimeng Gu***, Huidong Liu, Qiang Liu, Shu Wu, Haichao Shi, Xiao-Yu Zhang
In *ACL*, 2025 (Oral Presentation), *Equal Contributions
3. R²FND: Reinforced Rationale Learning for Fake News Detection with LLMs
Zhao Tong*, **Yimeng Gu***, Qiang Liu, Haichao Shi, Shu Wu, Xiao-Yu Zhang
In *IJCNN*, 2025, *Equal contribution
4. Contrastive Domain Adaptation with Test-Time Training for Out-of-Context News Detection
Yimeng Gu, Mengqi Zhang, Ignacio Castro, Shu Wu, Gareth Tyson
In *Pattern Recognition*, 2025
5. GatedVAE: Detecting Multimodal Fake News with Gated Variational AutoEncoder
Yimeng Gu, Ignacio Castro, Gareth Tyson
In *ACM Web Science*, 2024
6. MMVAE at SemEval-2022 Task 5: A Multi-modal Multi-task VAE on Misogynous Meme Detection
Yimeng Gu, Ignacio Castro, Gareth Tyson
In *NAACL SemEval Workshop*, 2022

HONORS AND AWARDS

- ACM WebSci'24 Travel Grant Apr 2024
- HKUST MSc (BDT) Excellent Student Scholarship 2019 (Top 7%) Apr 2020
- Excellent Undergraduate in Beihang University June 2018
- Second Prize in Science and Technology Competition in Beihang University (Top 15%) May 2017
- Honorable Mention in COMAP's Mathematical Contest in Modeling Apr 2016

LANGUAGE AND SKILLS

Computer/Technical	Python, Java, C/C++, SQL, Git, Gradle, CI/CD, Hadoop, Spark, AWS, Azure, GCP, VS Code, IntelliJ IDEA, PyCharm, AutoCAD
Deep Learning	Pytorch, Tensorflow, LLM Post-Training, LLM Evaluation, CUDA Prompt Engineering, RAG, LLM Agent, GenAI, HuggingFace, Wandb
Language	English (fluent), Mandarin (native)